226_Final_Project

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```
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v forcats   1.0.0     v readr     2.1.5
## v ggplot2   3.5.1     v stringr   1.5.1
## v lubridate 1.9.3     v tibble    3.2.1
## v purrr     1.0.2     v tidyr     1.3.1

## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(brms)
```

```
## Loading required package: Rcpp
## Loading 'brms' package (version 2.23.0). Useful instructions
## can be found by typing help('brms'). A more detailed introduction
## to the package is available through vignette('brms_overview').
##
## Attaching package: 'brms'
##
## The following object is masked from 'package:stats':
##
##     ar
```

```r
library(tidybayes)
```

```
##
## Attaching package: 'tidybayes'
##
## The following objects are masked from 'package:brms':
##
##     dstudent_t, pstudent_t, qstudent_t, rstudent_t
```

```r
library(bayesplot)
```

```
## Warning: package 'bayesplot' was built under R version 4.4.3
```

```
## This is bayesplot version 1.15.0
## - Online documentation and vignettes at mc-stan.org/bayesplot
## - bayesplot theme set to bayesplot::theme_default()
##    * Does _not_ affect other ggplot2 plots
##    * See ?bayesplot_theme_set for details on theme setting
##
## Attaching package: 'bayesplot'
##
## The following object is masked from 'package:brms':
##
##     rhat
```

1. Scientific Question

Do baseline tumor size (longest diameter, LD) and shape (sphericity), measured by MRI before treatment begins, predict pathologic complete response (pCR)? Does this relationship differ by molecular subtype defined by hormone receptor (HR) and HER2 status?

Understanding which pre-treatment tumor characteristics predict response could help clinicians stratify patients at diagnosis before any imaging follow-up is available.

Data Preparation

```r
clinical <- read_excel("~/Downloads/ISPY2-Imaging-Cohort-1-Clinical-Data.xlsx")
write.csv(clinical, "~/Downloads/ISPY2-Imaging-Cohort-1-Clinical-Data.csv", row.names = FALSE)

feature <- read_excel("~/Downloads/Multi-feature-MRI-NACT-Data.xlsx")
write.csv(feature, "~/Downloads/Multi-feature-MRI-NACT-Data.csv", row.names = FALSE)
```

convert xlsx file to csv file

```r
clinical <- read.csv("~/Downloads/ISPY2-Imaging-Cohort-1-Clinical-Data.csv")
feature <- read.csv("~/Downloads/Multi-feature-MRI-NACT-Data.csv")

#colnames(clinical)
#colnames(feature)

merged_df <- full_join(clinical, feature, by = c("Patient_ID" = "CLINICAL.TRIAL.SUBJECT.ID"))

write.csv(merged_df, "~/Downloads/226_fp_merged.csv", row.names = FALSE)
```

merged the two dataset The dataset was first examined for missing values in the outcome and predictor variables. The primary outcome variable pCR had no missing values. Some baseline imaging variables, including LD_T0 and SPHERICITY_T0, contained missing observations. To ensure consistent model fitting, I restricted the analysis to complete cases for the selected variables.

```r
df <- read.csv("~/Downloads/226_fp_merged.csv")
# Check missing values for each column
colSums(is.na(df))
```

```
##           Patient_ID                  Arm                   HR
##                    0                    0                    0
##                 HER2                   MP                  pCR
##                    0                    0                    0
##     Age_at_Screening                 Race    menopausal_status
##                    3                    2                   38
##            ethnicity   VOLUME_TUM_BLU_V10   VOLUME_TUM_BLU_V20
##                    1                  601                  601
##   VOLUME_TUM_BLU_V30   VOLUME_TUM_BLU_V40        SPHERICITY_T0
##                  601                  601                  601
##        SPHERICITY_T1        SPHERICITY_T2        SPHERICITY_T3
##                  601                  601                  601
##               LD_T0                LD_T1                LD_T2
##                  601                  601                  601
##               LD_T3   BPE_5slice_mean_T0   BPE_5slice_mean_T1
##                  601                  601                  601
##   BPE_5slice_mean_T2   BPE_5slice_mean_T3        FTV_pch_T0_T1
##                  601                  601                  601
##        FTV_pch_T0_T2        FTV_pch_T0_T3 Sphericity_pch_T0_T1
##                  601                  601                  601
## Sphericity_pch_T0_T2 Sphericity_pch_T0_T3         LD_pch_T0_T1
##                  601                  601                  601
##         LD_pch_T0_T2         LD_pch_T0_T3        BPE_pch_T0_T1
##                  601                  601                  601
##        BPE_pch_T0_T2        BPE_pch_T0_T3
##                  601                  601
```

```r
vars <- df %>%
  select(pCR, LD_T0, SPHERICITY_T0, HR, HER2)
colSums(is.na(vars))
```

```
##           pCR         LD_T0 SPHERICITY_T0            HR          HER2
##             0           601           601             0             0
```

```r
dat <- df %>%
  select(pCR, LD_T0, SPHERICITY_T0, HR, HER2) %>%
  drop_na()
nrow(dat)
```

```
## [1] 384
```

After removing missing values, 384 complete observations were included in the analysis.

```r
dat <- dat %>%
  mutate(
    pCR = as.integer(pCR),
    HR = factor(HR),
    HER2 = factor(HER2)
  )
str(dat)
```

```
## 'data.frame':    384 obs. of  5 variables:
##  $ pCR          : int  1 0 1 0 0 0 0 0 1 0 ...
##  $ LD_T0        : num  2.6 11.1 3 6 6.7 2.9 2.7 4.1 5.6 2.7 ...
##  $ SPHERICITY_T0: num  0.289 0.203 0.244 0.134 0.133 ...
##  $ HR           : Factor w/ 2 levels "0","1": 2 2 1 2 2 2 2 2 2 1 ...
##  $ HER2         : Factor w/ 2 levels "0","1": 2 1 1 1 1 1 2 1 1 1 ...
```

Continuous predictors (LD_T0 and SPHERICITY_T0) were standardized to have mean 0 and standard deviation 1 before modeling. This improves numerical stability of the Bayesian estimation process and allows prior distributions to be specified on a comparable scale across predictors.

```r
dat <- df %>%
  select(pCR, LD_T0, SPHERICITY_T0, HR, HER2) %>%
  drop_na() %>%
  mutate(
    pCR = as.integer(pCR),
    HR = factor(HR),
    HER2 = factor(HER2),
    LD_T0_z = as.numeric(scale(LD_T0)),
    SPHERICITY_T0_z = as.numeric(scale(SPHERICITY_T0))
  )
```

2. Model Specification

We modeled the probability of pathologic complete response (pCR) using Bayesian logistic regression. Because pCR is a binary outcome (yes/no), a Bernoulli likelihood with a logit link was used. The outcome for patient i was defined as:

$$pCR_i \sim \text{Bernoulli}(p_i)$$

with linear predictor

$$\text{logit}(p_i) = \alpha + \beta_1 \text{LD_T0_z}_i + \beta_2 \text{SPHERICITY_T0_z}_i + \beta_3 \text{HR}_i + \beta_4 \text{HER2}_i$$

where:

- α is the overall baseline log-odds of response
- β_1 is the effect of baseline lesion diameter
- β_2 is the effect of baseline sphericity
- β_3 is the effect of HR status
- β_4 is the effect of HER2 status

HR and HER2 were coded as binary predictors (0/1).

The Bayesian logistic regression model was implemented using the brms package in R.

```
#fit2 <- brm(
#  pCR ~ LD_T0_z + SPHERICITY_T0_z + HR + HER2,
#  data = dat,
#  family = bernoulli(),
#  prior =  priors2,
#  chains = 4,         # Number of MCMC chains
#  iter = 4000,        # Total iterations per chain
#  warmup = 2000,      # Burn-in iterations
#  cores = 4,
#  seed = 123
#)
```

3. Priors

Specify priors for the model parameters. Include at least two informative priors and briefly justify them.

```
get_prior(
  pCR ~ LD_T0_z + SPHERICITY_T0_z +
        HR + HER2,
  data = dat,
  family = bernoulli()
)
```

```
##                  prior     class            coef group resp dpar nlpar lb ub tag
##                 (flat)         b
##                 (flat)         b           HER21
##                 (flat)         b             HR1
##                 (flat)         b         LD_T0_z
##                 (flat)         b SPHERICITY_T0_z
##  student_t(3, 0, 2.5) Intercept
##        source
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
```

```
priors2 <- c(
  prior(normal(0, 1.5), class = "Intercept"),
  prior(normal(0, 0.7), class = "b", coef = "LD_T0_z"),
  prior(normal(0, 0.7), class = "b", coef = "SPHERICITY_T0_z"),
```

```
  prior(normal(0, 1), class = "b", coef = "HR1"),
  prior(normal(0, 1), class = "b", coef = "HER21")
)
```

- Intercept prior: normal(0, 1.5)
- This is a weakly informative prior on the log-odds scale. It says the baseline probability of pCR is uncertain, but extremely extreme values are less likely.
- LD_T0_z prior: normal(0, 0.7)
- This is informative. Since LD is standardized, a one-SD increase in tumor size is not expected to produce an absurdly huge change in log-odds. This prior allows moderate positive or negative effects while shrinking implausibly large effects.
- SPHERICITY_T0_z prior: normal(0, 0.7)
- Also informative. Sphericity may be associated with response, but the effect is not expected to be extremely large on the log-odds scale. This prior reflects moderate plausible effects.
- HR and HER2 priors: normal(0, 1)
- These are mildly informative priors. Molecular subtype may influence pCR, but the direction and size are uncertain, so these priors allow meaningful effects while still providing regularization.

```
fit2 <- brm(
  pCR ~ LD_T0_z + SPHERICITY_T0_z + HR + HER2,
  data = dat,
  family = bernoulli(),
  prior = priors2,
  chains = 4,
  iter = 4000,
  warmup = 2000,
  cores = 4,
  seed = 123
)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.0.13.3)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Library/Framew
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core
##   679 | #include <cmath>
##       |          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling
```

4. Posterior Results

Summarize the posterior results. Interpret results using short bullet points rather than long prose.

```
summary(fit2)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: pCR ~ LD_T0_z + SPHERICITY_T0_z + HR + HER2
##    Data: dat (Number of observations: 384)
##   Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -0.51      0.17    -0.84    -0.18 1.00    10091     6623
## LD_T0_z            -0.26      0.14    -0.56     0.01 1.00     8289     6600
## SPHERICITY_T0_z     0.13      0.13    -0.12     0.37 1.00     8495     6256
## HR1                -1.24      0.24    -1.72    -0.79 1.00     8150     5896
## HER21               0.95      0.26     0.44     1.46 1.00     8153     6151
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
fixef(fit2)
```

```
##                   Estimate Est.Error       Q2.5        Q97.5
## Intercept       -0.5083175 0.1697401 -0.8444215 -0.181254135
## LD_T0_z         -0.2628037 0.1441814 -0.5559557  0.009033271
## SPHERICITY_T0_z  0.1262780 0.1257920 -0.1243894  0.368845405
## HR1             -1.2404059 0.2366786 -1.7156178 -0.785038432
## HER21            0.9542795 0.2604876  0.4415933  1.464406197
```

```
exp(fixef(fit2))
```

```
##                  Estimate Est.Error      Q2.5     Q97.5
## Intercept       0.6015068  1.184997 0.4298059 0.8342233
## LD_T0_z         0.7688928  1.155094 0.5735239 1.0090742
## SPHERICITY_T0_z 1.1345975  1.134046 0.8830359 1.4460640
## HR1             0.2892668  1.267034 0.1798526 0.4561022
## HER21           2.5967989  1.297563 1.5551831 4.3249743
```

- A one standard deviation increase in baseline tumor size (LD_T0_z) was associated with lower odds of achieving pCR (OR = 0.77, 95% CrI: 0.57–1.01). The credible interval slightly overlapped 1, suggesting weak evidence of an association.
- A one standard deviation increase in tumor sphericity (SPHERICITY_T0_z) was associated with slightly higher odds of pCR (OR = 1.13, 95% CrI: 0.88–1.45). However, the credible interval includes 1, indicating no strong evidence of an association.
- Patients with HR-positive tumors had substantially lower odds of achieving pCR compared with HR-negative patients (OR = 0.29, 95% CrI: 0.18–0.46). The credible interval does not include 1, suggesting strong evidence of an association.

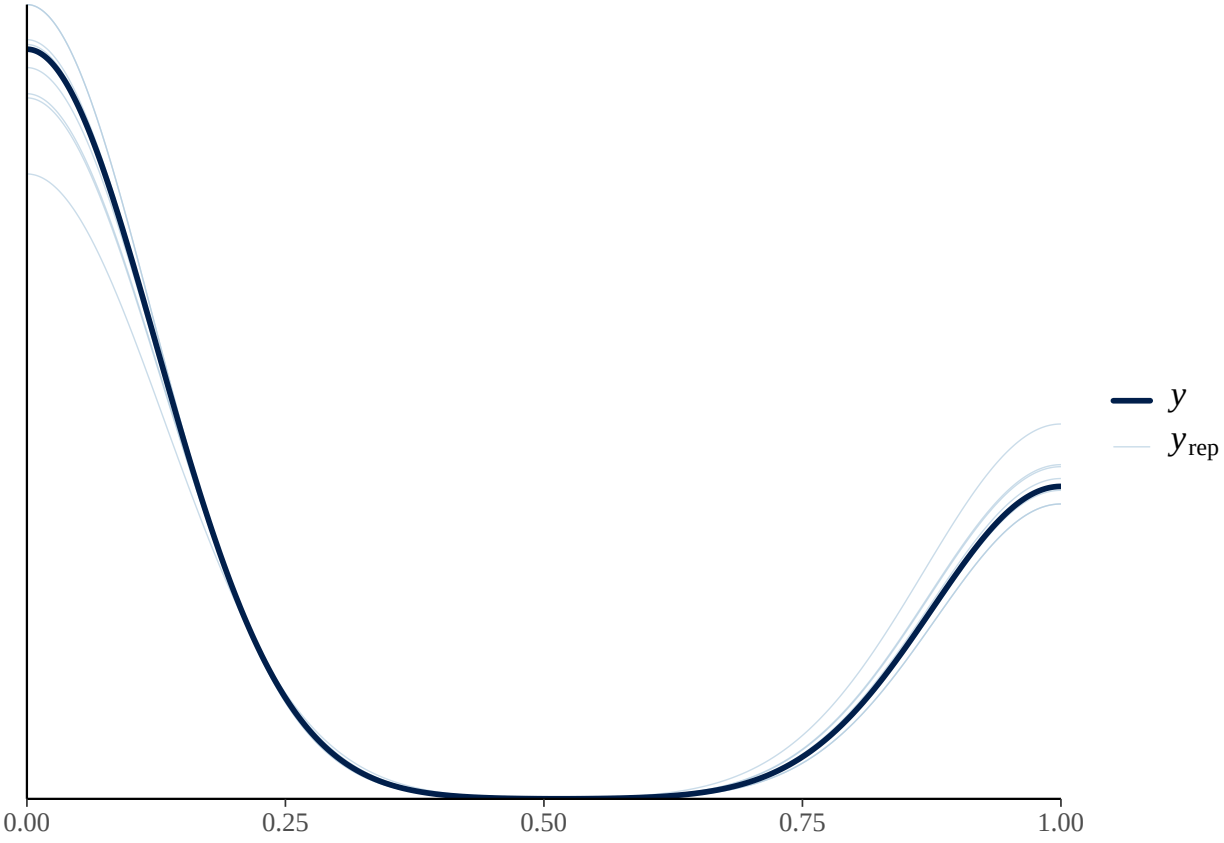
- Patients with HER2-positive tumors had significantly higher odds of achieving pCR compared with HER2-negative patients (OR = 2.60, 95% CrI: 1.56–4.32). The credible interval does not include 1, indicating strong evidence of a positive association.
- Overall, molecular subtype markers (HR and HER2 status) showed stronger associations with pCR than baseline imaging features (LD and sphericity).

5. Model Checking

Include appropriate posterior predictive checks or diagnostic plots.

```
pp_check(fit2)
```

```
## Using 10 posterior draws for ppc type 'dens_overlay' by default.
```



- The model reproduces the overall response pattern well, as the simulated distributions closely match the observed data.
- The variation is reproduced reasonably, with simulated values showing appropriate spread around the observed curve.
- No major discrepancies are observed, indicating adequate model fit.

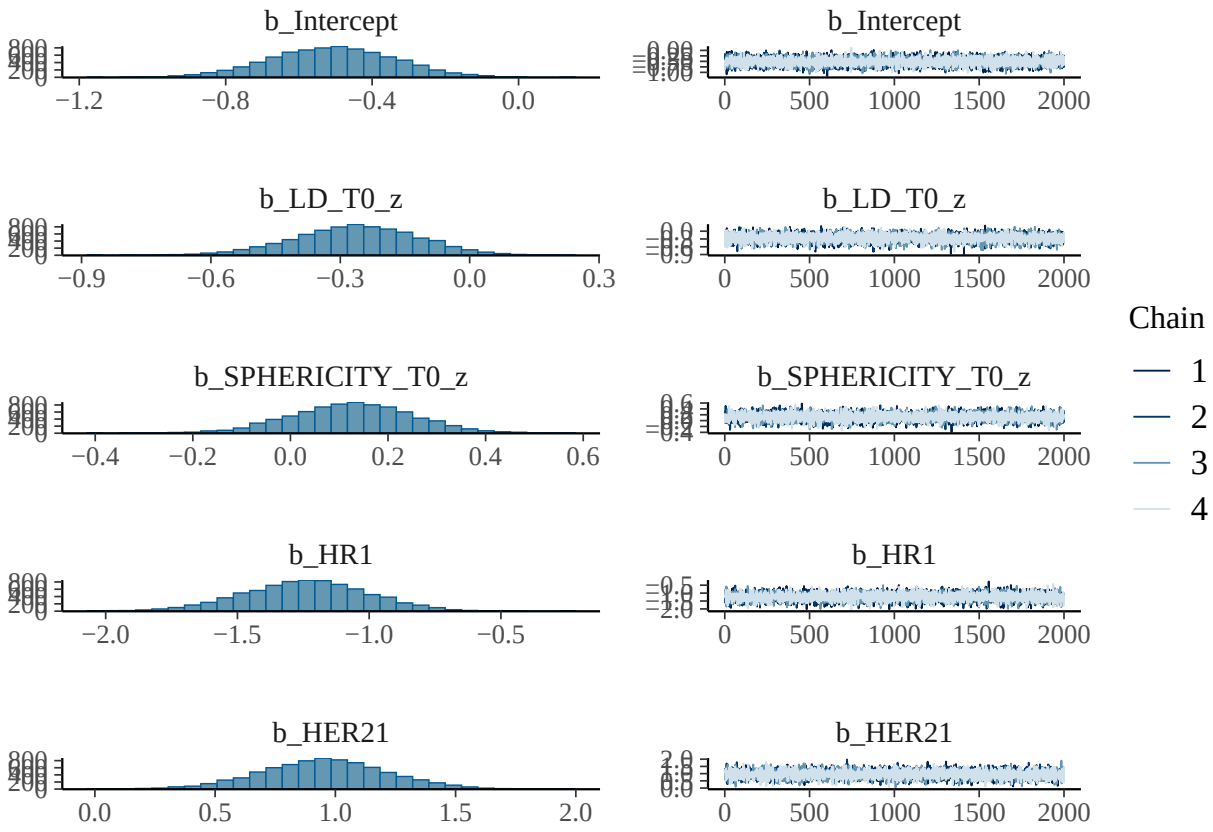
6. MCMC Diagnostics

Report convergence and mixing diagnostics (e.g., trace plots, Rˆ, effective sample size).

```
summary(fit2)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: pCR ~ LD_T0_z + SPHERICITY_T0_z + HR + HER2
##    Data: dat (Number of observations: 384)
##   Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -0.51      0.17    -0.84    -0.18 1.00    10091     6623
## LD_T0_z            -0.26      0.14    -0.56     0.01 1.00     8289     6600
## SPHERICITY_T0_z     0.13      0.13    -0.12     0.37 1.00     8495     6256
## HR1                -1.24      0.24    -1.72    -0.79 1.00     8150     5896
## HER21               0.95      0.26     0.44     1.46 1.00     8153     6151
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
plot(fit2)
```



- All parameters had R-hat values of 1.00, indicating good convergence.
- Effective sample sizes were large for all parameters, suggesting efficient posterior sampling.
- Trace plots showed good mixing, and no major Stan warning flags were observed, indicating satisfactory model diagnostics.

7. Model Comparison

Compare at least two models (for example using LOO or WAIC).

```
get_prior(
  pCR ~ LD_T0_z + SPHERICITY_T0_z,
  data = dat,
  family = bernoulli()
)
```

```
##                 prior     class            coef group resp dpar nlpar lb ub tag
##                (flat)         b
##                (flat)         b         LD_T0_z
##                (flat)         b SPHERICITY_T0_z
##  student_t(3, 0, 2.5) Intercept
##        source
##       default
##  (vectorized)
##  (vectorized)
##       default
```

```
get_prior(
  pCR ~ LD_T0_z + SPHERICITY_T0_z +
        HR * HER2,
  data = dat,
  family = bernoulli()
)
```

```
##                 prior     class            coef group resp dpar nlpar lb ub tag
##                (flat)         b
##                (flat)         b           HER21
##                (flat)         b             HR1
##                (flat)         b       HR1:HER21
##                (flat)         b         LD_T0_z
##                (flat)         b SPHERICITY_T0_z
##  student_t(3, 0, 2.5) Intercept
##        source
##       default
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##  (vectorized)
##       default
```

```r
# Priors for Model 1
priors1 <- c(
  prior(normal(0, 1.5), class = "Intercept"),
  prior(normal(0, 0.7), class = "b", coef = "LD_T0_z"),
  prior(normal(0, 0.7), class = "b", coef = "SPHERICITY_T0_z")
)

# Priors for Model 3
priors3 <- c(
  prior(normal(0, 1.5), class = "Intercept"),
  prior(normal(0, 0.7), class = "b", coef = "LD_T0_z"),
  prior(normal(0, 0.7), class = "b", coef = "SPHERICITY_T0_z"),
  prior(normal(0, 1), class = "b", coef = "HR1"),
  prior(normal(0, 1), class = "b", coef = "HER21"),
  prior(normal(0, 1), class = "b", coef = "HR1:HER21")
)
```

```r
# Fit Model 1
fit1 <- brm(
  pCR ~ LD_T0_z + SPHERICITY_T0_z,
  data = dat,
  family = bernoulli(),
  prior = priors1,
  chains = 4,
  iter = 4000,
  warmup = 2000,
  cores = 4,
  seed = 123
)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.0.13.3)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Library/Frame
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Cor
##   679 | #include <cmath>
##       |          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling
```

```r
# Fit Model 3
fit3 <- brm(
  pCR ~ LD_T0_z + SPHERICITY_T0_z + HR * HER2,
  data = dat,
  family = bernoulli(),
  prior = priors3,
  chains = 4,
  iter = 4000,
  warmup = 2000,
  cores = 4,
  seed = 123
)
```

```
## Compiling Stan program...
## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.0.13.3)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Library/Framew
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core
##   679 | #include <cmath>
##       |          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1
## in 1:
## Chain 3: Iteration: 3200 / 4000 [ 80%]  (Sampling)
## Chain 4: Iteration: 3200 / 4000 [ 80%]  (Sampling)
## Chain 2: Iteration: 3600 / 4000 [ 90%]  (Sampling)
## Chain 3: Iteration: 3600 / 4000 [ 90%]  (Sampling)
## Chain 4: Iteration: 3600 / 4000 [ 90%]  (Sampling)
## Chain 2: Iteration: 4000 / 4000 [100%]  (Sampling)
## Chain 2:
## Chain 2:  Elapsed Time: 0.058 seconds (Warm-up)
## Chain 2:                0.066 seconds (Sampling)
## Chain 2:                0.124 seconds (Total)
## Chain 2:
## Chain 4: Iteration: 4000 / 4000 [100%]  (Sampling)
## Chain 4:
## Chain 4:  Elapsed Time: 0.058 seconds (Warm-up)
## Chain 4:                0.061 seconds (Sampling)
## Chain 4:                0.119 seconds (Total)
## Chain 4:
## Chain 3: Iteration: 4000 / 4000 [100%]  (Sampling)
## Chain 3:
## Chain 3:  Elapsed Time: 0.057 seconds (Warm-up)
## Chain 3:                0.071 seconds (Sampling)
## Chain 3:                0.128 seconds (Total)
## Chain 3:
```

```
## Start sampling
```

```
summary(fit1)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: pCR ~ LD_T0_z + SPHERICITY_T0_z
##    Data: dat (Number of observations: 384)
##   Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -0.91      0.12    -1.14    -0.68 1.00     6431     5607
## LD_T0_z            -0.30      0.14    -0.58    -0.03 1.00     5651     5329
## SPHERICITY_T0_z     0.11      0.12    -0.13     0.34 1.00     6114     6043
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
summary(fit3)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: pCR ~ LD_T0_z + SPHERICITY_T0_z + HR * HER2
##    Data: dat (Number of observations: 384)
##   Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -0.53      0.18    -0.88    -0.19 1.00     8947     6208
## LD_T0_z            -0.27      0.14    -0.55    -0.00 1.00     8119     6170
## SPHERICITY_T0_z     0.13      0.12    -0.12     0.37 1.00     8549     6408
## HR1                -1.19      0.26    -1.71    -0.67 1.00     7588     6438
## HER21               1.08      0.37     0.38     1.81 1.00     5928     6050
## HR1:HER21          -0.21      0.47    -1.15     0.70 1.00     5347     5521
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```
loo1 <- loo(fit1)
loo2 <- loo(fit2)
loo3 <- loo(fit3)

#loo1
#loo2
#loo3

loo_compare(loo1, loo2, loo3)
```

```
##      elpd_diff se_diff
## fit2   0.0       0.0
## fit3  -0.5       0.4
## fit1 -17.9       5.8
```

The LOO comparison results were:

- Model 2 had the best predictive performance.
- Model 3 performed very similarly to Model 2, with an ELPD difference of -0.5 (SE = 0.4), indicating little to no meaningful improvement over Model 2.
- Model 1 performed substantially worse than the other two models, with an ELPD difference of -17.9 (SE = 5.8).

Therefore, Model 2 was selected as the preferred model, because it achieved the best expected out-of-sample predictive performance while being more parsimonious than Model 3.

8. Sensitivity Analysis

Refit the model using weak or default priors and comment on how the results change.

```
fit2_default <- brm(
  pCR ~ LD_T0_z + SPHERICITY_T0_z + HR + HER2,
  data = dat,
  family = bernoulli(),
  chains = 4,
  iter = 4000,
  warmup = 2000,
  cores = 4,
  seed = 123
)
```

```
## Compiling Stan program...

## Trying to compile a simple C file

## Running /Library/Frameworks/R.framework/Resources/bin/R CMD SHLIB foo.c
## using C compiler: 'Apple clang version 17.0.0 (clang-1700.0.13.3)'
## using SDK: ''
## clang -arch arm64 -I"/Library/Frameworks/R.framework/Resources/include" -DNDEBUG   -I"/Library/Framew
## In file included from <built-in>:1:
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/StanHeade
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## In file included from /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/
## /Library/Frameworks/R.framework/Versions/4.4-arm64/Resources/library/RcppEigen/include/Eigen/src/Core
##   679 | #include <cmath>
##       |          ^~~~~~
## 1 error generated.
## make: *** [foo.o] Error 1

## Start sampling
```

```r
summary(fit2_default)
```

```
##  Family: bernoulli
##   Links: mu = logit
## Formula: pCR ~ LD_T0_z + SPHERICITY_T0_z + HR + HER2
##    Data: dat (Number of observations: 384)
##   Draws: 4 chains, each with iter = 4000; warmup = 2000; thin = 1;
##          total post-warmup draws = 8000
##
## Regression Coefficients:
##                 Estimate Est.Error l-95% CI u-95% CI Rhat Bulk_ESS Tail_ESS
## Intercept          -0.50      0.17    -0.84    -0.16 1.00    10100     6349
## LD_T0_z            -0.28      0.15    -0.58     0.00 1.00     8687     6940
## SPHERICITY_T0_z     0.13      0.13    -0.13     0.37 1.00     8842     6941
## HR1                -1.33      0.25    -1.83    -0.85 1.00     8351     6198
## HER21               1.05      0.28     0.51     1.60 1.00     8947     6295
##
## Draws were sampled using sampling(NUTS). For each parameter, Bulk_ESS
## and Tail_ESS are effective sample size measures, and Rhat is the potential
## scale reduction factor on split chains (at convergence, Rhat = 1).
```

```r
exp(fixef(fit2_default))
```

```
##                  Estimate Est.Error      Q2.5     Q97.5
## Intercept       0.6090902  1.187214 0.4303055 0.8543232
## LD_T0_z         0.7578434  1.158087 0.5624777 1.0013666
## SPHERICITY_T0_z 1.1354733  1.135520 0.8798514 1.4507932
## HR1             0.2634606  1.283618 0.1611920 0.4277928
## HER21           2.8553754  1.320880 1.6676698 4.9540420
```

A sensitivity analysis was performed by refitting Model 2 using default (weak) priors instead of the original priors.

The results from the model with default priors were very similar to those from the original model. The effects of HR and HER2 stayed the same in both direction and size. HR-positive tumors were still associated with lower odds of achieving pCR, while HER2-positive tumors were associated with higher odds of achieving pCR.

Overall, these results suggest that the conclusions are stable and do not change much when different priors are used.

9. Conclusions

Provide a short summary of the key findings.

In this project, Bayesian logistic regression models were used to study factors associated with achieving pathological complete response (pCR). Several models were compared using leave-one-out cross-validation (LOO), and Model 2 was selected as the preferred model because it showed the best predictive performance while remaining relatively simple.

The results showed that HR status and HER2 status were important predictors of pCR. HR-positive tumors were associated with lower odds of achieving pCR, while HER2-positive tumors were associated with higher odds of achieving pCR. The imaging variables LD_T0_z and SPHERICITY_T0_z showed smaller effects and less consistent evidence of association.

A sensitivity analysis using default priors produced similar results to the original model, indicating that the conclusions were stable and not strongly influenced by prior choices.

Overall, the selected model provides useful insight into factors associated with treatment response and demonstrates the value of Bayesian methods for modeling clinical outcomes.